

Scalability & Future Plan

Date	1 July 2026
Team ID	XXXXXX
Project Name	HouseRent – House Rental Management System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the HouseRent – House Rental Management System can be enhanced to support a larger number of users, improve application performance, strengthen security, and introduce additional features. The roadmap focuses on ensuring that the system remains scalable, maintainable, and adaptable to future requirements.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Online payment functionality is not available.	Users cannot complete rental payments through the application.	High
2	Property location is displayed without map integration.	Users cannot easily locate properties on a map.	Medium
3	The application is currently optimized for small to medium user traffic.	Performance may decrease with a large number of concurrent users.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for limited concurrent users.	Deploy on cloud infrastructure with load balancing to support more users.
2	Data Storage	Uses MongoDB for local data storage.	Migrate to MongoDB Atlas with automated backups and scalable cloud storage.
3	Performance	Basic API optimization and database queries.	Implement caching, indexing, and optimized database queries to improve response time.
4	Security	JWT authentication and bcrypt password encryption are implemented.	Introduce Multi-Factor Authentication (MFA), HTTPS, and enhanced security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate online payment gateway and Google Maps location services.	Next Release	Improve convenience and user experience.
Phase 3	Add property reviews, ratings, email notifications, and live chat support.	Future Version	Increase user engagement and communication.
Phase 4	Implement AI-based property recommendations, mobile application support, and cloud deployment.	Long-Term	Enhance scalability, personalization, and accessibility.